

Vendor Management & Contingent Workforce Management System

1. Project Overview

The proposed system is a vendor-centric workforce management platform designed to manage the complete lifecycle of contractors working on client projects.

The platform connects three major stakeholders:

1. **Vendor**
2. **Contractor**
3. **Client / Product Manager**

The system manages:

- Client project requirements
- Contractor onboarding and document verification
- Contractor allocation
- Contractor availability
- Timesheet submission
- Timesheet approval
- Project milestones
- Billing calculation
- Invoice generation
- Invoice tracking

The main objective is to replace manual coordination between vendors, contractors and clients with a centralized workflow.

2. Problem Statement

In a traditional vendor-managed workforce process, project requirements, contractor allocation, working hours, approvals and billing may be handled through emails, spreadsheets and separate systems.

This creates problems such as:

- Difficulty tracking contractor assignments
- Accidental assignment of one contractor to multiple projects

- Manual verification of contractor documents
- Manual timesheet collection
- Delays in client approval
- Errors in calculating billable hours
- Manual invoice preparation
- Lack of visibility into project and billing status
- Difficulty maintaining an audit trail

The proposed system addresses these issues through a centralized vendor management platform.

3. Proposed Solution

The system provides a role-based platform where:

Client

↓

Project Requirements

↓

Vendor

↓

Contractor Verification

↓

Contractor Assignment

↓

Contractor Works

↓

Timesheet Submission

↓

Client Approval

↓

Milestone / Billing Calculation



Invoice Generation



Client

The vendor acts as the central entity responsible for managing projects and contractors.

4. User Roles

4.1 Vendor

The vendor is the primary administrator.

Responsibilities

- Manage clients
 - Create and manage projects
 - Define project requirements
 - Define project milestones
 - Add contractors
 - Verify contractor documents
 - Check contractor availability
 - Assign contractors
 - Monitor timesheets
 - Track milestone progress
 - Calculate billing
 - Generate invoices
 - Track invoice status
-

4.2 Contractor

The contractor performs work for the assigned project.

Responsibilities

- Login to the system
- View assigned projects
- View project requirements
- View assigned role
- Enter working hours
- Submit timesheets
- View timesheet status
- Correct rejected timesheets
- Resubmit timesheets

Contractors cannot:

- Assign themselves to projects
 - Modify project billing
 - Approve their own timesheets
 - Generate invoices
-

4.3 Client / Product Manager

The client-side Product Manager validates the contractor's work.

Responsibilities

- View assigned projects
 - View assigned contractors
 - Review submitted timesheets
 - Approve contractor hours
 - Reject incorrect hours
 - Provide rejection comments
 - View project milestones
 - View invoices
-

5. Core Modules

The system will contain the following modules:

1. Authentication & Authorization
2. Client Management
3. Project Management
4. Contractor Management
5. Contractor Verification
6. Contractor Assignment
7. Timesheet Management
8. Timesheet Approval
9. Milestone Management
10. Billing Management
11. Invoice Management
12. Dashboard & Reporting
13. Audit Logging

For the 7-hour MVP, the highest-priority modules are:

Authentication

Project Management

Contractor Management

Contractor Assignment

Timesheets

Approval

Milestones

Billing

Invoices

6. End-to-End Business Workflow

Step 1 — Client Requirement

The client provides a project requirement to the vendor.

Example:

Client:

ABC Bank

Project:

Banking Application

Required Skills:

Java

Spring Boot

React

PostgreSQL

Required Contractors:

2 Java Developers

1 React Developer

1 QA Engineer

Step 2 — Vendor Creates Project

Vendor enters:

- Project name
- Client
- Description
- Requirements
- Start date
- End date
- Budget
- Milestones

Step 3 — Contractor Onboarding

Vendor adds contractors.

Contractor profile contains:

- Name
- Email
- Phone
- Skills
- Experience
- Documents
- Verification status

Step 4 — Contractor Verification

Contractor documents are uploaded.

Example:

Government ID

PAN

Qualification Certificate

Experience Certificate

Verification status:

PENDING

VERIFIED

REJECTED

Only a verified contractor can be assigned to a project.

7. Contractor Assignment

The vendor selects:

Project

Contractor

Role

Start Date

End Date

Hourly Rate

Before creating the assignment, the backend performs two checks.

Check 1 — Verification

Contractor verification status = VERIFIED

If not:

Assignment rejected.

Contractor is not verified.

Check 2 — Assignment Conflict

The system checks whether the contractor already has an assignment during the selected period.

Example:

Existing Assignment

Project A

01-Aug → 30-Sep

Vendor attempts:

Project B

15-Sep → 15-Oct

The system detects an overlap.

 Assignment Conflict

Contractor is already assigned

to Project A during this period.

This prevents double allocation.

8. Timesheet Management

Once assigned, the contractor records working hours.

Example:

Week: 10-Aug → 16-Aug

Monday 8 hours

Tuesday 8 hours

Wednesday 7 hours

Thursday 8 hours

Friday 8 hours

Total 39 hours

Timesheet status:

DRAFT

SUBMITTED

APPROVED

REJECTED

9. Timesheet Approval

After submission:

Contractor

↓

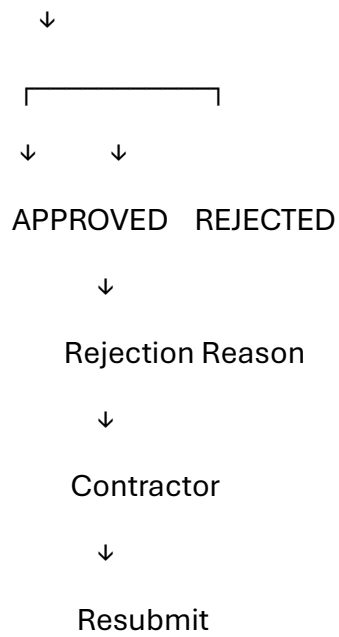
SUBMITTED

↓

Client / Product Manager

↓

Review



Only approved timesheets should become eligible for billing.

10. Milestone Management

Each project can contain multiple milestones.

Example:

Project: Banking Application

Milestone 1

Requirement Analysis

₹50,000

Milestone 2

Backend Development

₹2,00,000

Milestone 3

Frontend Development

₹1,50,000

Milestone 4

Testing

₹75,000

Milestone statuses:

PENDING

IN_PROGRESS

COMPLETED

BILLED

11. Billing

The system will calculate billing based on approved work.

For hourly billing:

Billing Amount =

Approved Hours × Hourly Rate

Example:

Approved Hours = 40

Hourly Rate = ₹1,000

Billing Amount = $40 \times 1,000$

= ₹40,000

For milestone billing:

Milestone Completed

+

Required Work Approved

↓

Billing Eligible

The vendor can generate an invoice only when the billing conditions are satisfied.

12. Invoice Generation

Invoice generation is performed by the vendor.

Example:

Invoice Number:

INV-2026-001

Vendor:

XYZ Technologies

Client:

ABC Bank

Project:

Banking Application

Milestone:

Backend Development

Approved Hours:

195

Subtotal:

₹2,00,000

Tax:

₹36,000

Total:

₹2,36,000

Status:

GENERATED

Invoice statuses:

GENERATED

SENT

PAID

For the 7-hour MVP, payment gateway integration is not required.

13. Dashboard

Vendor Dashboard

Display:

Active Projects

Active Contractors

Pending Timesheets

Pending Invoices

Upcoming Milestones

Example:

Vendor Dashboard

Active Projects 5

Active Contractors 18

Pending Timesheets 7

Pending Invoices 2

Projects

Banking Application Active

HR Platform Active

CRM Platform Billing

Contractor Dashboard

Display:

Assigned Projects

Pending Timesheets

Approved Hours

Rejected Timesheets

Client Dashboard

Display:

Active Projects

Pending Timesheets

Approved Timesheets

Pending Invoice Review

14. Technology Stack

Frontend

React.js

Tailwind CSS

React Router

Axios

Backend

Node.js

Express.js

JWT

bcrypt

Database

MySQL

Sequelize ORM

Development Tools

VS Code

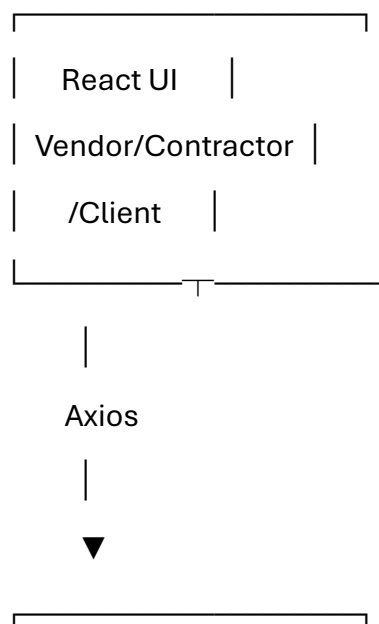
Git

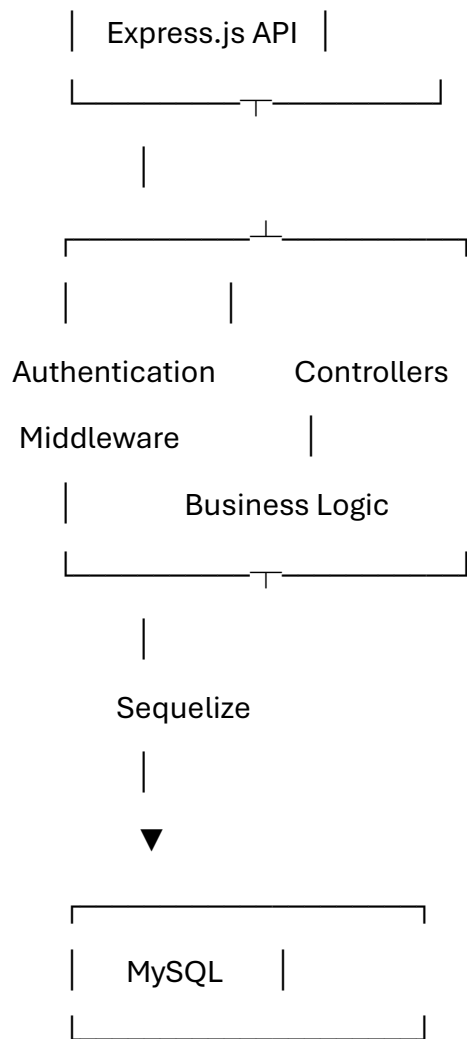
GitHub

Postman

MySQL Workbench

15. System Architecture





16. MySQL Database Design

Users

users

id

name

email

password_hash

role

created_at

updated_at

Roles:

VENDOR

CONTRACTOR

CLIENT

Vendors

vendors

id

user_id

company_name

contact_number

address

created_at

Clients

clients

id

user_id

company_name

contact_number

address

created_at

Contractors

contractors

id
user_id
phone
experience
verification_status
created_at

Contractor Documents

contractor_documents

id
contractor_id
document_type
document_url
verification_status
verified_by
verified_at
created_at

Projects

projects

id
vendor_id
client_id
name
description
start_date

end_date
budget
status
created_at

Project Requirements

project_requirements

id
project_id
skill
required_count
description

Project Milestones

project_milestones

id
project_id
name
description
amount
start_date
end_date
status

Project Assignments

project_assignments

id
project_id
contractor_id
role
start_date
end_date
hourly_rate
status
created_at

Timesheets

timesheets

id
contractor_id
project_id
week_start
week_end
total_hours
status
submitted_at
approved_by
approved_at
rejection_reason

Timesheet Entries

timesheet_entries

id
timesheet_id
work_date
hours
description

Invoices

invoices

id
invoice_number
vendor_id
client_id
project_id
milestone_id
subtotal
tax
total
status
generated_at
sent_at
paid_at

Invoice Items

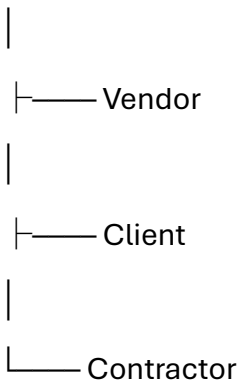
invoice_items

id

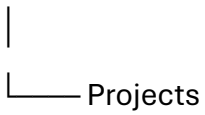
invoice_id
description
quantity
rate
amount

17. Database Relationships

User



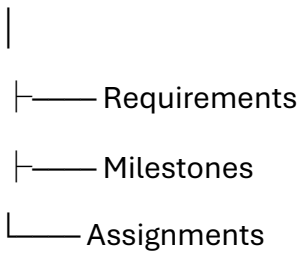
Vendor



Client



Project



Contractor

|

|— Documents

|— Assignments

|— Timesheets

Timesheet

|

|— Timesheet Entries

Project + Milestone + Approved Work

|

|— Invoice

|

|— Invoice Items

18. API Design

Authentication

POST /api/auth/login

POST /api/auth/register

Projects

POST /api/projects

GET /api/projects

GET /api/projects/:id

PUT /api/projects/:id

DELETE /api/projects/:id

Contractors

POST /api/contractors

GET /api/contractors

GET /api/contractors/:id

PUT /api/contractors/:id

PUT /api/contractors/:id/verify

Assignments

POST /api/assignments

GET /api/assignments

DELETE /api/assignments/:id

Timesheets

POST /api/timesheets

GET /api/timesheets

GET /api/timesheets/pending

PUT /api/timesheets/:id/approve

PUT /api/timesheets/:id/reject

Milestones

POST /api/milestones

GET /api/milestones/:projectId

PUT /api/milestones/:id

Invoices

POST /api/invoices/generate

GET /api/invoices

GET /api/invoices/:id

PUT /api/invoices/:id/send

19. Role-Based Authorization

Feature	Vendor	Contractor	Client
Login	✓	✓	✓
Create Project	✓	-	-
View Project	✓	✓	✓
Add Contractor	✓	-	-
Verify Contractor	✓	-	-
Assign Contractor	✓	-	-
Submit Timesheet	-	✓	-
Approve Timesheet	-	-	✓
Reject Timesheet	-	-	✓
Manage Milestones	✓	-	View
Generate Invoice	✓	-	-
View Invoice	✓	-	✓

20. Frontend Pages

Authentication

Login

Vendor

Vendor Dashboard

Projects

Project Details

Contractors

Contractor Details

Assignments

Timesheets

Milestones

Invoices

Contractor

Contractor Dashboard

My Projects

Project Details

Submit Timesheet

Timesheet History

Client

Client Dashboard

Projects

Pending Timesheets

Timesheet Review

Invoices

21. Backend Folder Structure

server/

|

├─ config/

|

└─ database.js

|

├─ models/

|

├─ User.js

|

├─ Vendor.js

|

└─ Client.js

- | | └─ Contractor.js
- | | └─ ContractorDocument.js
- | | └─ Project.js
- | | └─ ProjectRequirement.js
- | | └─ ProjectMilestone.js
- | | └─ ProjectAssignment.js
- | | └─ Timesheet.js
- | | └─ TimesheetEntry.js
- | | └─ Invoice.js
- | | └─ InvoiceItem.js
- |
- | └─ controllers/
 - | | └─ authController.js
 - | | └─ projectController.js
 - | | └─ contractorController.js
 - | | └─ assignmentController.js
 - | | └─ timesheetController.js
 - | | └─ milestoneController.js
 - | | └─ invoiceController.js
- |
- | └─ routes/
 - | | └─ authRoutes.js
 - | | └─ projectRoutes.js
 - | | └─ contractorRoutes.js
 - | | └─ assignmentRoutes.js
 - | | └─ timesheetRoutes.js
 - | | └─ milestoneRoutes.js

```
|   └─ invoiceRoutes.js
|
|   └─ middleware/
|       └─ authMiddleware.js
|           └─ roleMiddleware.js
|
|   └─ services/
|       └─ billingService.js
|           └─ invoiceService.js
|
└─ server.js
```

22. Frontend Folder Structure

client/

```
|
|
└─ src/
    |
    |   └─ components/
    |       └─ Sidebar.jsx
    |       └─ Navbar.jsx
    |       └─ StatusBadge.jsx
    |       └─ DataTable.jsx
    |           └─ Modal.jsx
    |
    |   └─ pages/
    |       └─ Login.jsx
    |
    |
```

```
|   ├── vendor/
|   |   ├── Dashboard.jsx
|   |   ├── Projects.jsx
|   |   ├── Contractors.jsx
|   |   ├── Assignments.jsx
|   |   ├── Timesheets.jsx
|   |   ├── Milestones.jsx
|   |   └── Invoices.jsx
|   |
|   ├── contractor/
|   |   ├── Dashboard.jsx
|   |   ├── Projects.jsx
|   |   ├── SubmitTimesheet.jsx
|   |   └── TimesheetHistory.jsx
|   |
|   └── client/
|       ├── Dashboard.jsx
|       ├── Timesheets.jsx
|       └── Invoices.jsx
|
|── services/
|   └── api.js
|
|── context/
|   └── AuthContext.jsx
|
|── App.jsx
```

└─ main.jsx

23. 7-Hour Implementation Plan

Hour 1 — Setup + Database + Authentication

Tasks

- Create React project
- Create Express server
- Configure MySQL
- Configure Sequelize
- Create User model
- Create authentication API
- Implement JWT
- Implement role middleware
- Test login using Postman

Deliverable

Vendor Login

Contractor Login

Client Login

Hour 2 — Vendor + Project + Contractor

Tasks

- Create Vendor Dashboard
- Create Project API
- Create Project UI
- Create Contractor API
- Create Contractor UI
- Add contractor verification

Deliverable

Vendor

↓

Create Project

↓

Add Contractor

↓

Verify Contractor

Hour 3 — Assignment

Tasks

- Assignment model
- Assignment API
- Assignment UI
- Verification check
- Date validation
- Overlap detection

Critical feature

Existing Assignment

01-Aug → 30-Sep

New Assignment

15-Sep → 15-Oct

Result:

 Conflict

Hour 4 — Contractor Timesheets

Tasks

- Contractor dashboard
- Project list
- Timesheet form
- Timesheet API
- Timesheet history
- Submit functionality

Deliverable

Contractor

↓

View Project

↓

Enter Hours

↓

Submit

Hour 5 — Client Approval

Tasks

- Client dashboard
- Pending timesheet page
- Timesheet details
- Approve
- Reject
- Rejection reason

Deliverable

Contractor

↓

Submit

↓

Client



Approve / Reject

Hour 6 — Milestone + Billing + Invoice

Tasks

- Milestone creation
- Milestone status
- Approved hours calculation
- Billing calculation
- Invoice generation
- Invoice display

Deliverable

Approved Timesheet



Milestone



Billing



Invoice

Hour 7 — Integration + Testing + Demo

Do not add new functionality during the final hour.

Test the complete workflow:

Vendor Login



Create Project



Create Milestone



Add Contractor



Verify Contractor



Assign Contractor



Contractor Login



Submit Timesheet



Client Login



Approve Timesheet



Vendor Login



Complete Milestone



Generate Invoice

Test:

- Login
- Role authorization
- Project creation
- Contractor verification
- Assignment conflict
- Timesheet submission

- Timesheet approval
 - Timesheet rejection
 - Billing calculation
 - Invoice generation
-

24. Priority Classification

Because the development time is only 7 hours, features should be divided into priorities.

P0 — Must Work

Authentication

Project creation

Contractor creation

Contractor verification

Contractor assignment

Assignment conflict detection

Timesheet submission

Timesheet approval

Milestone

Invoice generation

P1 — If Time Is Available

Document upload

Invoice sending

Dashboard statistics

Search

Filtering

Audit logs

P2 — Future Development

Payment gateway

Email notifications

Advanced identity verification

OCR

AI contractor matching

Advanced analytics

Payroll integration

ERP integration

Mobile application

25. Important Business Rules

Rule 1

Only verified contractors can be assigned.

IF contractor.verification_status != VERIFIED

THEN reject assignment

Rule 2

Contractor cannot have overlapping assignments.

new_start < existing_end

AND

new_end > existing_start

Rule 3

Only contractors can submit their own timesheets.

Rule 4

Only clients can approve/reject submitted timesheets.

Rule 5

Only approved hours are billable.

Rule 6

Rejected timesheets must contain a rejection reason.

Rule 7

Invoice should be generated only from valid billing data.

Rule 8

Once an invoice is generated, its amount should not silently change when a timesheet is modified.

26. Transaction Handling

Financial operations should use database transactions.

For invoice generation:

BEGIN TRANSACTION

1. Check project
2. Check milestone
3. Check milestone status
4. Fetch approved timesheets
5. Calculate amount
6. Create invoice
7. Create invoice items
8. Update milestone → BILLED

COMMIT

If an operation fails:

ROLLBACK

This protects invoice consistency.

27. Demo Scenario

Use one complete scenario during presentation.

Vendor

TechVendor Pvt Ltd

Client

ABC Bank

Project

Banking Application

Contractor

Arun

Role: Java Developer

Rate: ₹1,000/hour

Milestone

Backend Development

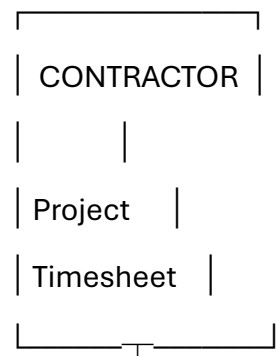
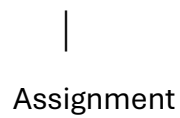
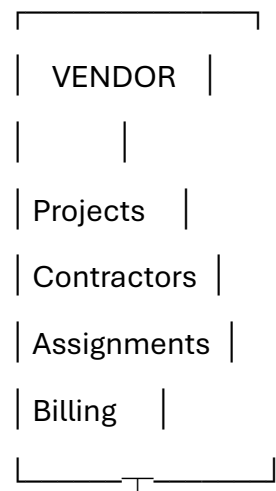
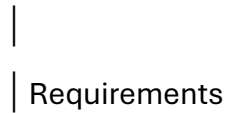
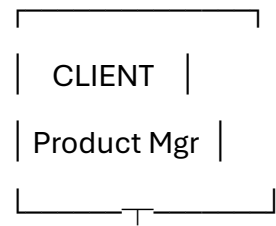
Amount: ₹2,00,000

Demo

1. Vendor logs in
2. Creates Banking Application
3. Adds Arun
4. Uploads/marks documents as verified
5. Assigns Arun to project
6. System checks availability
7. Arun logs in
8. Arun submits 40 hours
9. Client logs in
10. Client approves 40 hours
11. Vendor sees approved hours
12. Vendor completes milestone
13. System calculates billing
14. Vendor generates invoice
15. Invoice is displayed

This demonstrates the entire business lifecycle.

28. Final MVP Architecture



Submit Hours

|



CLIENT	
Approve	
Reject	

|

Approved Hours

|



VENDOR	
Milestone	
Billing	
Invoice	

29. Final Technology Decision

The final implementation stack is:

Frontend:

React + Tailwind CSS

Backend:

Node.js + Express.js

Database:

MySQL

ORM:

Sequelize

Authentication:

JWT + bcrypt

API:

REST

Testing:

Postman

Version Control:

Git + GitHub

The project should be implemented as a **single modular application**, not microservices, because the objective is to produce a working MVP within 7 hours.

30. Project Definition

Short Description

Vendor Management & Contingent Workforce Management System is a web-based platform that enables vendors to manage client projects, verify and allocate contractors, track contractor working hours, obtain client approvals, manage project milestones, calculate billable work and generate invoices.

Core Value Proposition

Verified Workforce

+

Conflict-Free Allocation

+

Timesheet Approval

+

Milestone Tracking

+

Automated Billing

+

Invoice Generation

This transforms the system from a simple timesheet tracker into a **complete vendor-centric contingent workforce management MVP**.